

Effects of different conditioning programmes on the performance of high-velocity soccer-related tasks: Systematic review and meta-analysis of controlled trials

Amador García-Ramos^{1,4}, Guy Gregory Haff², Belén Feriche¹
and Slobodan Jaric³

Abstract

Physical performance is a key factor in most of the actions that determine success in soccer competitions. The present meta-analysis aimed to determine whether, and to what extent different conditioning programmes implemented in addition to regular soccer training are able to enhance the performance of high-velocity soccer-related tasks. Study groups were categorised according to type of performance measurement: (A) vertical jump ability, (B) linear sprint performance, (C), change of direction with ball, (D) change of direction without ball, (E) repeated sprint ability, and (F) kicking performance. Subgroup analyses were done with reference to the type of conditioning programme applied: (A) resistance training, (B) plyometric, (C) sprint and/or change of direction, and (D) combined methods. The standardised mean differences (SMD) along with 95% confidence intervals (95% CI) were calculated using the Review Manager 5.3 software. The application of a conditioning programme was associated with a significant improvement in vertical jump ability (SMD = 0.64 [95% CI: 0.50–0.77]), linear sprint (SMD = –0.79 [95% CI: –0.61 to –0.96]), change of direction with ball (SMD = –0.78 [95% CI: –0.42 to –1.15]), change of direction without ball (SMD = –0.83 [95% CI: –0.61 to –1.05]), repeated sprint ability (SMD = –0.51 [95% CI: –0.19 to –0.84]), and kicking performance (SMD = 0.78 [95% CI: 0.34–1.22]). No clear differences among the different conditioning programmes were observed. The implementation of a conditioning programme in addition to regular soccer training enhances the performance of high-velocity soccer-related tasks.

Keywords

Agility, Association football, change of direction, kicking performance, plyometric training, repeated sprint ability, resistance training, strength and conditioning, vertical jump

Introduction

The latest large-scale FIFA survey conducted in 2007 leaves no doubt that soccer is the most practiced sport with 265 million players worldwide.¹ The high level of competence needed to become a professional soccer player as well as the slight difference in quality among top soccer teams requires the optimisation of all factors related to soccer performance, such as the technical, tactical, physical, physiological, and psychological.² One of the main characteristics that discriminates top soccer players (e.g. Lionel Messi and Cristiano Ronaldo; recognised as the best soccer players from 2008 to 2015) is their ability to perform soccer-specific actions at high speed.² In this regard, scoring goals during soccer competitions are typically preceded by

Reviewer: Lee Brown (California State University, Fullerton, USA)

¹Faculty of Sport Sciences, Department of Physical Education and Sport, University of Granada, Granada, Spain

²Center for Exercise and Sport Science Research, Edith Cowan University, Joondalup, Australia

³Department of Kinesiology and Applied Physiology & Biomechanics, Movement Science Graduate Program, University of Delaware, Newark, DE, USA

⁴Universidad Católica de la Santísima Concepción, Facultad de Educación, Concepción, Chile

Corresponding author:

Amador García-Ramos, Faculty of Sport Sciences, Department of Physical Education and Sport, University of Granada, Ctra. de Alfacar, Granada 18011, Spain.

Email: amagr@ugr.es

at least one powerful action such as straight sprints, jumps, and/or changes of direction.³

During competition, soccer players are required to repeatedly produce maximal or near-maximal efforts (e.g. accelerations, sprints, change of direction, jumps, and kicks) interspersed with brief recovery intervals.⁴ On the basis of the undeniable importance of the actions described above, training studies have been conducted with the expectation of improving the performance of these high-velocity actions through the application of different conditioning programmes. Resistance training,^{5–16} plyometric training,^{5,9,17–31} sprint or changes of direction training,^{18,32–40} and different combinations of these methods^{5,15,41–55} have been the conditioning programmes most used to elucidate their effectiveness to enhance the performance of high-velocity soccer-related tasks (e.g. vertical jump, linear sprint, change of direction with and without ball, repeated sprint ability, and kicking velocity). These training methods are often combined with traditional soccer training and then compared to soccer training alone.

The recent systematic review conducted by Silva et al.⁵⁶ has suggested that neuromuscular training can enhance both physiological and physical measures related with the performance of soccer players. However, to date, to the authors' knowledge, no study has quantified the combined effects of the different conditioning programmes that have been applied in conjunction with regular soccer training on high-velocity tasks related to soccer performance (e.g. jumps, sprints, changes of direction, repeated sprint ability, and kicking performance). Therefore, the findings of this meta-analysis are expected to provide soccer professionals with a more detailed estimation of the effects of adding a conditioning training programme to regular soccer training in comparison to perform soccer training alone on physical performance in soccer players.

The present meta-analysis aimed to determine whether, and to what extent different conditioning programmes implemented in addition to regular soccer training are able to enhance the performance of high-velocity soccer-related tasks. We hypothesised that the implementation of conditioning programmes would induce higher increases in performance than only practising soccer.

Methods

Literature search

The US National Library of Medicine (PubMed), MEDLINE, Google Scholar, and SportDiscus databases were searched up to July 2016 for research articles investigating the effect of conditioning training on physical performance in soccer players. The search was performed using the following keywords

combinations in the English and Spanish languages: 'soccer', 'football', 'resistance training', 'strength', 'power', 'plyometric', 'weight-lifting', 'concurrent training', 'jump', 'sprint', 'agility', 'change of direction', and 'kicking performance'. Additionally, the reference lists and citations of the identified studies were explored using Google Scholar to find additional articles. Attempts were also made to contact the authors of the selected articles to request any missing relevant information. The study protocol adhered to the tenets of the Declaration of Helsinki and was approved by the Institutional Review Board.

Inclusion and exclusion criteria

Studies were chosen if they fulfilled the following six inclusion criteria: (i) the studied athletic population consisted of soccer players older than 14 years, (ii) the experimental group was compared against a similar group that only performed soccer training (control group), (iii) the training programmes applied in the experimental groups were detailed and supervised, (iv) the training programmes consisted on (A) resistance training, (B) plyometric training, (B) sprint or change of direction training, or (D) a combination among at least two of these categories, (v) they included at least one of the following dependent variables: vertical jump ability, linear sprint, agility or change of direction with or without a ball, repeated sprint ability, and kicking performance, and (vi) the article was written in the English or Spanish languages and published as an original article in a peer-reviewed journal.

Studies were excluded if soccer players used supplementation, cycling, electrostimulation, or vibration training methods. The studies were also excluded from the sprint training category if the subjects did not perform all-out sprints (maximum possible running intensity), or if a treadmill was used for training. After critically analysing the initial studies collected with the above criteria, a cohort of 51 studies was selected for the systematic review. However, seven of these studies were excluded from the meta-analysis because they failed to report the mean and standard deviation of the dependent variables of interest needed to calculate the effect sizes (Figure 1).

Data extraction and quality assessment

The included studies were coded according to the participant's characteristics (age and sex), phase of the season of the intervention (pre-season or in-season), duration of the intervention, number of conditioning training sessions per week, and type of conditioning programme applied. Information was archived independently in a database by two authors and

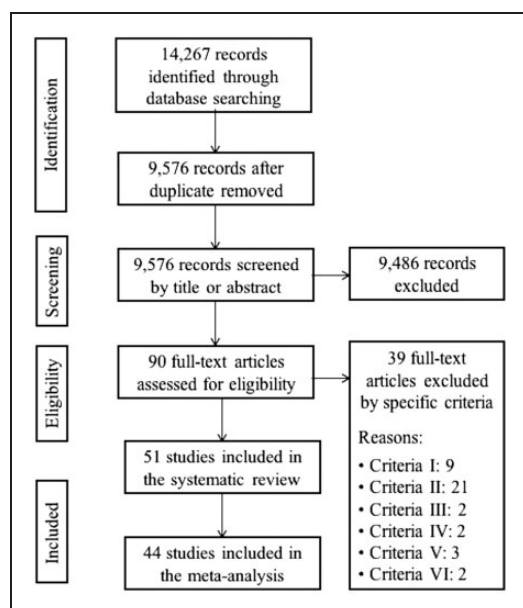


Figure 1. Flow diagram of the studies that underwent the review process. *Criteria I:* No control group; *Criteria II:* Soccer players younger than 14 years; *Criteria III:* Sprint training no performed at maximal intensity; *Criteria IV:* Do not measure any of the dependent variable of the meta-analysis; *Criteria V:* Training or testing procedures no specified; *Criteria VI:* Training in off-season.

discrepancies were resolved by consensus. The quality of the 44 studies included in the meta-analysis was evaluated according to the Physiotherapy Evidence Database (PEDro) scale.⁵⁷ However, the blinding of participants and therapists who administered the training was not possible due to the characteristics of the studies, so the maximum number of points possible was 9.

Categorisation of interventions

The general characteristics of the four conditioning programmes considered in the present systematic review is provided below. The specific characteristics of each of the studies included in these categories can be found in Table 1.

- A. *Resistance Training.* The conditioning programmes consisted of a variety of resistance training exercises such as the back squat, leg press, lunge, calf raise, or deadlift. The number of sets per exercise, repetitions per set, and intensity (generally expressed as a percentage of the one-repetition maximum [1RM]) of each study is reported in Table 1.
- B. *Plyometric training.* The conditioning programmes consisted of a variety of vertical and/or horizontal body weight jumping-type exercises using the stretch-shortening cycle muscle action in which

soccer players were instructed to minimise ground contact time and to maximise jumping height (in vertical jumps) or distance (in horizontal jumps).

- C. *Sprint or change of direction training.* The conditioning programmes consisted of maximal sprints either in a straight line or with changes of direction.
- D. *Combination of methods.* This category includes the experimental groups that used at least two of the training methods described above.

Types of outcomes measure

- A. *Vertical jump ability.* The ability to perform powerful vertical jumps can be decisive in important soccer actions such as the headed shot. Vertical jump height was the variable used to determine vertical jump ability. The meta-analysis of vertical jump ability has included different unloaded vertical jumps such as the countermovement jump (CMJ), Abalakov, and the vertical jump after a previous race. Not to overestimate the precision of the studies with multiple outcomes for the vertical jump category, only one of the reported outcomes has been included in the meta-analysis. The CMJ was selected when multiple outcomes were reported because it was the jump-type most reported in the studies included in the meta-analysis.
- B. *Linear sprint performance.* Goals and other decisive actions in soccer are often preceded by accelerations and sprints.^{3,33} Sprint time (in s) or maximum velocity (in m/s) were the variables used to determine the linear sprint performance. A better sprint performance is associated with a shorter time or a higher velocity. To deal with this issue, we changed the direction of the maximum velocity mean change before conducting the analysis. The meta-analysis of linear sprint performance included different sprint times from 10 to 40 m as well as the maximum sprint velocity. To avoid overestimation of the precision of the studies with multiple outcomes for the linear sprint category, only one of the reported outcomes has been included in the meta-analysis. The time to 20 m was chosen whenever possible because this was the sprint distance most assessed in the studies included in the present meta-analysis. If the distance to 20 m was not assessed, we selected the distance closest to 20 m among the different sprint distances evaluated. The shortest distance was selected in case of equality (i.e. 10 m instead of 30 m).
- C. *Change of direction with ball.* The ability to perform quick and unpredictable changes of direction with the ball is important in order to dribble opponents and to progress towards the opposing goal.

Table 1. General characteristics of the studies included in the systematic review.

Study name	Intervention	Subjects	Protocol	Response
Brito et al. ^{5a}	RT	CG: 21 men; 21 ± 1 years EG: 12 men; 20 ± 1 years	Nine weeks of in-season training. 2 RT × week One set of six repetitions at 85–90% of 1RM in the squat, calf extension, and leg extension exercises	= vertical jump ↑ linear sprint = agility without ball
Chelly et al. ^{6a}	RT	CG: 11 men; 17 ± 0.5 years EG: 11 men; 17 ± 0.3 years	Eight weeks of in-season training. 2 RT × week Seven repetitions at 70%RM, four repetitions at 80%RM, three repetitions at 85%RM, and two repetitions at 90%RM in the back half squat exercise.	↑ vertical jump ↑ linear sprint
De Hoyo et al. ^{7a}	RT	CG: 15 men; 17 ± 1 years EG: 18 men; 18 ± 1 years	Ten weeks of in-season training. 1–2 RT × week 3–6 sets of 6 repetitions in the half squat and leg curl performed in a YoYo isoinertial flywheel training device.	↑ vertical jump ↑ linear sprint
Enoksen et al. ^{8a}	RT	CG: 9 men; 19 ± 3.5 years EG: 9 men; 19 ± 3.5 years	Ten weeks of pre-season training. 2 RT × week Week 1–6 the programme focused on hypertrophy: three sets of 6–12 repetitions at 70–85%RM. Week 7–10 the programme focused on maximal strength: 3–4 sets of 2–6 repetitions at 80–95%RM. The key exercises were the back squat, leg press, and dead lift	= vertical jump ↑ linear sprint
Haghighi et al. ^{9a}	RT	CG: 10 men; 19 ± 1.5 years EG: 10 men; 18 ± 0.8 years	Eight weeks of training 2–4 sets of 6–12 repetitions at 60–90%RM in the leg extension, leg curls, leg press, and seated calf raise exercises	↑ linear sprint ↑ agility with ball
Jelusic et al. ^{10a}	RT	CG: 10 men; 16 years EG: 10 men; 16 years	15 weeks of in-season training. 2 RT × week Three sets of six repetitions with a weight of 7 kg in a pulley device simulating the real soccer kick	↑ kicking velocity
Jullien et al. ¹¹	RT	CG: 9 men; 19 ± 1.7 years EG: 9 men; 18 ± 1.6 years	Three weeks of pre-season training. 5 RT × week Three sets of three repetitions at 90%RM in the squat exercise	= linear sprint = agility with ball = repeated sprint
Karsten et al. ^{12a}	RT	CG: 13 men; 19 ± 1 years EG: 13 men; 18 ± 1 years	Six weeks of pre-season training. 2 RT × week Three sets of 12 repetitions at 60–65%RM (week 1–2) and 3–4 sets of 70–75%RM (week 3–6) in the parallel squat, lunges, and Romanian deadlift exercises	↑ linear sprint
Manolopoulos et al. ^{13a}	RT	CG: 8 men; 21 ± 1.3 years EG: 8 men; 21 ± 1.3 years	Eight weeks of pre-season training. 3 RT × week Week 1–2: 3–4 sets of 20 repetitions at 50–60%RM in the leg extension, leg curl, adductor machine, calf machine, and leg press exercises. Simulated soccer kicks using a pulley device. Week 3–4: 4–6 sets of 6–8 repetitions at 70–80%RM in the same exercises. Week 5–6: 3–5 sets of an upward pyramid and 3–6 sets of 1–5 repetitions at 75–95%RM	= linear sprint ↑ kicking velocity

(continued)

Table 1. Continued

Study name	Intervention	Subjects	Protocol	Response
Manolopoulos et al. ¹⁴	RT	CG: 10 men; 21 ± 1.3 years EG: 10 men; 20 ± 1.4 years	Ten weeks of pre-season training. 3 RT × week Week 1–2: 3 sets of 10 repetitions at 50%RM in exercises that targeted the upper and lower body limbs. Week 3–5: 3 sets of 10 repetitions at 70%RM in 6 exercises. Week 6–8: 5 sets of 2–6 repetitions at 70–90%RM in 6 exercises. Week 9–10: 4 sets of 1–4 repetitions at 80–95%RM	↑ kicking velocity
Rønnestad et al. ^{15a}	RT	CG: 7 men; 24 ± 1.5 years EG: 6 men; 23 ± 2.0 years	Six weeks of pre-season training. 2 RT × week 3–5 sets of 4–6 repetitions in the half squat and hip flexion exercises	= vertical jump = linear sprint
Sanchez-Sanchez et al. ^{16a}	RT	CG: 18 men; 14 ± 0.2 years EG: 20 men; 15 ± 0.4 years	Ten weeks of in-season training. 2 RT × week 3–5 sets of 6–12 repetitions at 60–75%RM in 5 specific exercises that involved the calves, quadriceps, hamstrings, adductors, and abductors muscles	= linear sprint
Brito et al. ^{5a}	PT	CG: 21 men; 21 ± 1 years EG: 12 men; 20 ± 1 years	Nine weeks of in-season training. 2 PT × week Three stations of skipping, maximal vertical jumps, and drop jumps from 60 cm	= vertical jump ↑ linear sprint = agility without ball
Chelly et al. ^{17a}	PT	CG: 11 men; 19 ± 0.8 years EG: 12 men; 19 ± 0.7 years	Eight weeks of in-season training. 2 PT × week 4–10 sets of 10 repetitions in the 40/60-cm hurdle jump or 40-cm drop jump exercises.	↑ vertical jump ↑ linear sprint
Chettiamkudiyil et al. ¹⁸	PT	CG: 10 men; 18–26 years EG: 10 men; 18–26 years	Six weeks of in-season training. 2–3 PT × week Low, medium, and high intensity exercises such as double leg hop, scissors jump, single leg hop, medicine ball throw in single hand, etc.	↑ vertical jump ↑ linear sprint
Fábrica et al. ^{19a}	PT	CG: 20 men; 25 ± 3.1 years EG: 20 men; 25 ± 3.1 years	Six weeks of training. 3 PT × week Coordinative variants of multi-jumps, vertical jumps, and horizontal jumps. Subjects performed 500–1000 jumps per week	↑ vertical jump
Gauffin et al. ^{20a}	PT	CG: 16 men; 20 ± 3 years EG: 31 men; 20 ± 3 years	Ten weeks of pre-season training. 3 PT × week Three sets of 10 maximum vertical jumps	↑ vertical jump
Haghighi et al. ^{9a}	PT	CG: 10 men; 19 ± 1.5 years EG: 10 men; 19 ± 1.7 years	Eight weeks of training The exercises performed were the depth jump, the split squat jump, the rim jump, the box to box depth jump	↑ linear sprint ↑ agility with ball
Hammami et al. ^{21a}	PT	CG: 13 men; 16 ± 0.2 years EG: 15 men; 16 ± 0.2 years	Eight weeks of in-season training. 2 PT × week Five-week 1–4: 10 sets of 7–10 repetitions in the 0.5–0.6 m hurdle jump; Week 5–8: 4 sets of 7–10 repetitions in the 0.6–0.7 m drop jump	↑ linear sprint = agility without ball = repeated sprint

(continued)

Table 1. Continued

Study name	Intervention	Subjects	Protocol	Response
Kumar ^{22a}	PT	CG: 15 men; 20–25 years EG: 15 men; 20–25 years	Five weeks of training. 3 PT × week. Plyometric exercises of low (e.g. diagonal cone hops), medium (e.g. double leg hops), and high intensity (e.g. lateral jump single leg)	↑ agility without ball
Kumar ^{23a}	PT	CG: 15 men; 20–25 years EG: 15 men; 20–25 years	Six weeks of training. 3 PT × week. Plyometric exercises of low (e.g. diagonal cone hops), medium (e.g. double leg hops), and high intensity (e.g. lateral jump single leg)	↑ vertical jump
Memarzadeh et al. ²⁴	PT	CG: 10 men; 22 ± 0.5 years EG: 10 men; 22 ± 0.5 years	Eight weeks of training. The exercises performed were the depth jump, the split squat jump, the rim jump, the box to box depth jump	↑ linear sprint ↑ agility with ball ↑ agility without ball
Ozbar ^{25a}	PT	CG: 10 women; 19 ± 1.7 years EG: 10 women; 19 ± 1.6 years	Ten weeks of in-season training. 2 PT × week 3–5 sets of 5–8 repetitions in 6–8 exercises, with a minimum of 120 and maximum 250 foot contact per training session	↑ vertical jump ↑ linear sprint ↑ kicking velocity
Ozbar et al. ^{26a}	PT	CG: 9 women; 18 ± 2.0 years EG: 9 women; 18 ± 2.6 years	Eight weeks of in-season training. 1 PT × week 4–5 sets of 5–15 repetitions in 4–5 exercises, with a minimum of 90 and maximum 220 foot contact per training session	↑ vertical jump ↑ linear sprint
Ramirez-Campillo et al. ^{28a}	PT	CG: 10 women; 23 ± 2.1 years EG: 10 women; 23 ± 1.7 years	Six weeks of in-season training. 2 PT × week Unilateral and bilateral horizontal and vertical jumps with both cyclic and acyclic arm swings	= vertical jump ↑ linear sprint = agility without ball = repeated sprint
Ramirez-Campillo et al. ^{28a}	PT	CG: 21 men; 21 ± 2.7 years EG: 21 men; 20 ± 2.8 years	Six weeks of in-season training. 2 PT × week Two sets of 5–10 repetitions in 12 jump exercises (i.e. cyclic and acyclic horizontal and vertical jumps, with left, right and both legs)	↑ vertical jump ↑ linear sprint ↑ agility without ball
Ramirez-Campillo et al. ^{28a}	PT	CG: 19 women; 21 ± 2.5 years EG: 19 women; 22 ± 2.4 years	Six weeks of in-season training. 2 PT × week Two sets of 5–10 repetitions in 12 jump exercises (i.e. cyclic and acyclic horizontal and vertical jumps, with left, right and both legs)	↑ vertical jump ↑ linear sprint ↑ agility without ball
Sedano et al. ^{29a}	PT	CG: 10 women; 23 ± 3.2 years EG: 10 women; 23 ± 2.1 years	Twelve weeks of in-season training. 3 PT × week Jumps over hurdles of 60 cm, drop jumps from 50 cm, and horizontal jumps	↑ vertical jump ↑ kicking velocity
Sedano et al. ^{30a}	PT	CG: 11 women; 18 ± 0.9 years EG: 11 women; 18 ± 1.1 years	Ten weeks of in-season training. 3 PT × week Jumps over hurdles of 60 cm, horizontal jumps, and lateral jumps over hurdles. 180–290 jumps per week	↑ vertical jump ↑ linear sprint ↑ kicking velocity

(continued)

Table 1. Continued

Study name	Intervention	Subjects	Protocol	Response
Vaczi et al. ^{31a}	PT	CG: 12 men; 23 ± 1.4 years EG: 12 men; 22 ± 1.7 years	Six weeks of in-season training. 2 PT × week Double-leg hurdle jump (3–6 sets of 5 repetitions), single-leg lateral jump (2–4 sets of 10 repetitions), single-leg forward hop (2–4 sets of 5 repetitions), double-leg depth jump (2–6 sets of 5 repetitions), double-leg lateral jump (2–6 sets of 5 repetitions), and single-leg hurdle jump (2–4 sets of 10 repetitions).	↑ vertical jump ↑ agility without ball
Chaouachi et al. ^{32a}	ST	CG: 12 men; 14 ± 0.9 years EG: 12 men; 14 ± 0.9 years	Six weeks of in-season training. 3 ST × week Preplanned change of direction drills such as the Shuttle 4 × 10 m, Half-T-test 20 m, COD 5-0-5 m, and Skipping 10 m	↑ vertical jump ↑ linear sprint ↑ agility with ball ↑ agility without ball
Chettiamkudiyil et al. ¹⁸	ST	CG: 10 men; 18–26 years EG: 10 men; 18–26 years	Six weeks of in-season training. 2–3 ST × week Low, medium, and high intensity exercises such as s-drill, high knee forward, zigzag run, scissors jump, ladder speed run, etc.	↑ vertical jump = linear sprint
Dupont et al. ^{33a}	ST	CG: 22 men; 20 ± 0.7 years EG: 22 men; 20 ± 0.7 years	Ten weeks of in-season training. 2 ST × week Session 1: 12–15 sprints of 40 m with 30 seconds of passive recovery. Session 2: 2 sets of 12–15 intermittent runs of 15 seconds at 120% of maximal aerobic speed alternated with 15 s of passive recovery	↑ linear sprint
Jovanovic et al. ³⁴	ST	CG: 50 men; 19 years EG: 50 men; 19 years	Eight weeks of in-season training. 3 ST × week. Speed, agility, quickness (SAQ) training method	↑ vertical jump ↑ linear sprint
Kavaliuskas et al. ^{35a}	ST	CG: 7 men; 23 ± 7 years EG: 7 men; 22 ± 8 years	Six weeks of in-season training. 2 ST × week. 10 × 10 second sprints, on a gradient of 7%, followed by a 60-s recovery period that involved walking back to the start	↑ agility without ball
Mathisen et al. ^{36a}	ST	CG: 9 women; 15 ± 0.5 years EG: 10 women; 16 ± 0.7 years	Eight weeks of pre-season training. 1 ST × week The programme consisted of eight partner-resisted sprints (15 m), eight 20-m straight sprints, eight change of direction sprints (20 m), and finished with eight relay race with 90° turns	↑ linear sprint ↑ agility without ball
Nedrehagen et al. ^{37a}	ST	CG: 10 Mixed sex; 22 ± 2.6 years EG: 12 Mixed sex; 20 ± 3 years	Eight weeks of in-season training. 1 ST × week 3–4 sets of 4–6 repetitions of 20 + 10 m shuttle runs	↑ repeated sprint
Polman et al. ^{38a}	ST	CG: 12 women; 21 ± 3.1 years EG: 12 women; 21 ± 3.1 years	Twelve weeks of in-season training. 2 ST × week Speed, agility, quickness (SAQ) training method with various items of specialised resistance and speed development equipment	↑ vertical jump ↑ linear sprint ↑ agility without ball ↑ repeated sprint

(continued)

Table 1. Continued

Study name	Intervention	Subjects	Protocol	Response
Polman et al. ^{38a}	ST	CG: 12 women; 21 ± 3.1 years EG: 12 women; 21 ± 3.1 years	Twelve weeks of in-season training. 2 ST × week Speed, agility, quickness (SAQ) training method without the use of any specialised equipment	↑ vertical jump ↑ linear sprint ↑ agility without ball ↑ repeated sprint
Shalfawi et al. ^{39a}	ST	CG: 7 men; 16 ± 0.5 years EG: 8 men; 16 ± 0.5 years	Eight weeks of pre-season training. 2 ST × week Four sets of 5 × 40 m with 90-s recovery between repetitions and 10-min recovery between sets	= vertical jump ↑ linear sprint = agility without ball ↑ repeated sprint
Tønnessen et al. ^{40a}	ST	CG: 10 men; 16 ± 0.9 years EG: 10 men; 16 ± 0.9 years	Ten weeks of pre-season training. 1 ST × week 2–4 sets of 4–5 repetitions of 40 m sprint	= vertical jump = linear sprint ↑ repeated sprint
Brito et al. ^{5a}	CT	CG: 21 men; 21 ± 0.8 years EG: 12 men; 20 ± 1 years	Nine weeks of in-season training. 2 CT × week One set of six repetitions at 85–90% of IRM in the squat, calf extension, and leg extension exercises + 3 stations of skipping, maximal vertical jumps, and drop jumps from 60 cm	= vertical jump ↑ linear sprint = agility without ball
Cavaco et al. ^{41a}	CT	CG: 6 men; 14 ± 0.8 years EG: 5 men; 14 ± 0.5 years	Six weeks of pre-season training. 1 CT × week Two stations: (1) three sets of six repetitions at 85% of IRM in squat exercise + 15 m plus top speed and cross; (2) three sets of six repetitions at 85% of IRM in squat exercise + agility with the ball and shot at the goal	= linear sprint = agility with ball
Cavaco et al. ^{41a}	CT	CG: 6 men; 14 ± 0.8 years EG: 5 men; 14 ± 0.5 years	Six weeks of pre-season training. 2 CT × week Two stations: (1) three sets of six repetitions at 85% of IRM in squat exercise + 15 m plus top speed and cross; (2) three sets of six repetitions at 85% of IRM in squat exercise + agility with the ball and shot at the goal	= linear sprint = agility with ball
Faude et al. ^{42a}	CT	CG: 8 men; 23 ± 2.4 years EG: 8 men; 23 ± 2.7 years	Seven weeks of in-season training. 2 CT × week Session 1: four sets of four repetitions at 90%RM in the unilateral loaded half squats + 5 single leg hurdle jumps. Session 2: 5–8 repetitions at 50–60%RM in half squat, calf raises, lateral half squats, or step-ups followed by sprint, plyometric, or agility exercises	↑ vertical jump = linear sprint = agility with ball
Franco-Márquez et al. ^{43a}	CT	CG: 18 men; 15 ± 0.5 years EG: 20 men; 15 ± 0.5 years	Six weeks of in-season training. 2 CT × week 2–3 sets of 4–8 repetitions at 45–58%RM in the full squat exercise followed by vertical jumps, sprints, step phase triple jumps, and changes of direction	↑ vertical jump ↑ linear sprint

(continued)

Table 1. Continued

Study name	Intervention	Subjects	Protocol	Response
García-Pinillos et al. ^{44a}	CT	CG: 13 men; 16 ± 0.5 years EG: 17 men; 16 ± 0.5 years	Twelve weeks of in-season training. 2 CT × week. 4–6 sets of 1 repetition of 40–80 s in the 90° isometric half squat exercise followed by 6 repetitions in the jump from the seated position or single-leg jump exercises	↑ vertical jump ↑ linear sprint ↑ agility without ball ↑ kicking velocity
González-Badillo et al. ^{45a}	CT	CG: 11 men; 19 ± 1.2 years EG: 16 men; 18 ± 0.4 years	Twenty-six weeks of in-season training. 2 CT × week 2–4 sets of 4–8 repetitions at 80–105% of the 1 m/s load in the full squat exercise followed by exercises such as loaded countermovement jump, jump to box, step phase triple jumps, and accelerations with changes of direction	= vertical jump = linear sprint
González-Badillo et al. ^{45a}	CT	CG: 11 men; 19 ± 1.2 years EG: 17 men; 15 ± 0.3 years	Twenty-six weeks of in-season training. 2 CT × week 2–4 sets of 4–8 repetitions at 80–105% of the 1 m/s load in the full squat exercise followed by exercises such as loaded countermovement jump, jump to box, step phase triple jumps, and accelerations with changes of direction	↑ vertical jump = linear sprint
Gorostiaga et al. ⁴⁶	CT	CG: 11 men; 17 ± 0.7 years EG: 8 men; 17 ± 0.5 years	Eleven weeks of in-season training. 2 CT × week The main exercises were the full squat (2–3 sets of 2–6 repetitions), power clean (3–4 sets of 3–4 repetitions), vertical countermovement jump to box (3–5 sets of 5–8 repetitions), hurdle vertical jumps (three sets of 4 repetitions), and sprint runs (one set of 2–5 repetitions of 15–40 m)	↑ vertical jump = linear sprint
Hernández et al. ^{47a}	CT	CG: 27 men; 17 ± 0.7 years EG: 22 men; 18 ± 0.9 years	Eight weeks of in-season training. 2 CT × week Four sets of five repetitions with the load that maximises power output in the power clean, half squat, and loaded jump squat exercises, and three sets of 20 repetitions in the jump to a 40–50 cm box	↑ linear sprint
Hernández et al. ^{48a}	CT	CG: 27 men; 17 ± 0.7 years EG: 22 men; 18 ± 0.9 years	Eight weeks of in-season training. 2 CT × week Four sets of five repetitions with the load that maximises power output in the power clean, half squat, and loaded jump squat exercises, and three sets of 20 repetitions in the jump to a 40–50 cm box	↑ agility without ball
López-Segovia et al. ^{49a}	CT	CG: 12 men; 18 ± 0.8 years EG: 13 men; 18 ± 0.6 years	Sixteen weeks of in-season training. 2 CT × week 2–8 sets of 4–6 repetitions at 80–120% of the 1 m/s load in the full squat exercise followed by exercises such as the half squat, loaded and unloaded countermovement jump, sled towing, and step phase triple jump	= vertical jump ↓ linear sprint

(continued)

Table 1. Continued

Study name	Intervention	Subjects	Protocol	Response
Maio Alves et al. ^{50a}	CT	CG: 6 men; 17 ± 0.6 years EG: 9 men; 17 ± 0.6 years	Six weeks of pre-season training. 1 CT × week First station: six repetitions of the squat exercise at 85%RM followed by 1 set of 5 meters high skipping and 5 meters sprint. Second station: six repetitions of the calf extension exercise at 90%IRM followed by eight vertical jumps and three high ball headers. Third station: six repetitions of leg extension exercise at 80%RM followed by six jumps from the seated position and three drops jumps from 60 cm	= vertical jump ↑ linear sprint = agility without ball
Maio Alves et al. ^{50a}	CT	CG: 6 men; 17 ± 0.6 years EG: 8 men; 17 ± 0.6 years	Six weeks of pre-season training. 2 CT × week First station: six repetitions of the squat exercise at 85%RM followed by 1 set of 5 m high skipping and 5 m sprint. Second station: six repetitions of the calf extension exercise at 90%IRM followed by eight vertical jumps and three high ball headers. Third station: six repetitions of leg extension exercise at 80%RM followed by six jumps from the seated position and three drop jumps from 60 cm	= vertical jump ↑ linear sprint = agility without ball
Manolopoulos et al. ⁵¹	CT	CG: 10 men; 22 ± 1.3 years EG: 10 men; 20 ± 0.4 years	Ten weeks of training. 3 CT × week Week 1–2: three sets of 15–20 repetitions to improve general strength. Week 3–4: three sets of six repetitions of exercises such as skipping, jumping on one and both legs, and jumping running forwards, backwards, and to the side. Week 5–10: soccer-specific strength of the lower limbs using rubber band or another player as external resistance	↑ linear sprint ↑ kicking velocity
Rønnestad et al. ^{15a}	CT	CG: 7 men; 24 ± 1.5 years EG: 8 men; 23 ± 2.0 years	Seven weeks of pre-season training. 2 CT × week 3–5 sets of 4–6 repetitions in the half squat and hip flexion exercises + 2–4 sets of 5–10 repetitions in the double-arm single-leg forward jumps, single-arm alternate-leg forward bounce, and double-leg hurdle jumps	= vertical jump = linear sprint
Sáez de Villarreal et al. ^{52a}	CT	CG: 13 men; 15 ± 0.2 years EG: 13 men; 15 ± 0.3 years	Nine weeks of in-season training. 2 CT × week Half squat with jump (2–4 sets of 6–10 repetitions), skipping (2–4 sets of 10–20 m), stride length (2–4 sets of 6–10 repetitions), sidelong jumps of 30-cm hurdle (2–4 sets of 6–10 repetitions), vertical jumps in 30-cm hurdle (2–4 sets of 6–10 repetitions), and second support of triple jump (2–4 sets of 6–10 repetitions); 10-m sprint + technical dribbling + shoot were added to all exercises in each repetition	↑ vertical jump ↑ linear sprint ↑ agility with ball ↑ agility without ball ↑ kicking velocity

(continued)

Table 1. Continued

Study name	Intervention	Subjects	Protocol	Response
Siegler et al. ^{53a}	CT	CG: 17 women; 16 ± 1.4 years EG: 17 women; 16 ± 0.9 years	Ten weeks of in-season training. 5 CT × week Two sessions per week: three sets of four lower-body free-weight exercises (squats, leg extensions, leg curls, and calf raise). Three sessions per week: 3–5 sets of box jumps, repeated jumping, and skipping	= vertical jump ↑ linear sprint
Singh et al. ^{54a}	CT	CG: 14 men; professional players EG: 14 men; professional players	Six weeks of pre-season training. 3 CT × week RT: four sets at 6RM in the high-pull, jump squat, bench press, back half squat and chin-up exercises. PT: 80–120 foot contacts per session using the counter-movement jump and depth jump exercise	↑ vertical jump ↑ linear sprint ↑ agility without ball ↑ kicking velocity
Wong et al. ^{55a}	CT	CG: 19 men; 21 ± 1.0 years EG: 20 men; 25 ± 1.5 years	Eight weeks of pre-season training. 2 CT × week RT: four sets at 6RM in the high-pull, jump squat, bench press, back half squat and chin-up exercises. ST: 16 sets of 15 s of running at high speed	↑ vertical jump ↑ linear sprint =kicking velocity

RT: resistance training; PT: plyometric training; ST: Sprint or change of direction training; CT: combined training; CG: control group; EG: experimental group; ↑: net treatment effect significantly favoured the EG; =: net treatment effect did not significantly differ between CG and EG; ↓: net treatment effect significantly favoured the CG.

^aStudy included in the meta-analysis.

The time needed to complete a pre-planned change of direction protocol with the ball was the variable employed to determine this ability. Only one study reported multiple outcomes to evaluate the change of direction with ball category.⁵² Not to overestimate the precision of the study conducted by Sáez de Villarreal et al.,⁵² we only considered the protocol with the higher time at pre-test.

D. *Change of direction without ball.* Soccer players need to perform multiple changes of direction during competitive matches to cope the tactical demands of the game.² The time needed to complete a pre-planned change of direction protocol without the ball was the variable employed to determine this ability. The meta-analysis of change of direction without ball performance has included different pre-planned change of direction protocols such as the 10-8-8-10 test,³² 5-10-5 agility test,³⁵ Balsom agility test,⁴⁴ or Illinois test.^{27,54} To avoid overestimation of the precision of the studies with multiple outcomes for the change of direction without ball category, only the protocol with the highest time at pretest was included in the meta-analysis.

E. *Repeated sprint ability.* During matches, soccer players are required to repeatedly perform high-speed actions and sprints interspersed with brief

recovery periods.^{4,58} The average or total time needed to complete a repeated sprint ability protocol was the variable included in the meta-analysis to evaluate the repeated sprint ability. Neither study included in the meta-analysis reported more than one outcome of the tested repeated sprint ability.

F. *Kicking performance.* The ability to kick the ball faster increases the chances of scoring a goal during a soccer match.¹⁴ Ball speed (in m/s or km/h) was the variable used to determine kicking performance. Not to overestimate the precision of the studies with multiple outcomes for the kicking performance category, only one of the reported outcomes has been included in the meta-analysis. We selected the kicking velocity with the dominant leg as the main criterion. If the dominant leg was not reported, we selected the kicking velocity with the right leg.

Statistical analyses

Dependent variables data were extracted in the forms of pre- and post-training intervention means, standard deviations (SDs), and sample sizes for experimental and control conditions. Effect sizes were calculated

by subtracting the pre-intervention mean value from the post-intervention mean value (post-pre) for both the experimental ($\Delta 1$) and control groups ($\Delta 2$). The net treatment effect size was obtained as $\Delta 1$ minus $\Delta 2$ divided by the pooled SD of baseline values.⁵⁹ In studies that reported intermediate and post-intervention values, only final values were compared against baseline. In studies that reported more than one outcome to evaluate the same physical ability, only one of the outcomes (selection criteria are reported in section 3.5) was considered to avoid overestimation of the precision of the studies with multiple outcomes for the same ability.

Data were analysed using the Review Manager 5.3 software provided by the Cochrane Collaboration (<http://tech.cochrane.org/revman>). Random effects models were performed on each dependent variable using the inverse-variance method. Pooled estimates of the effect size obtained by either comprehensive or subgroup meta-analyses are reported as standardised mean differences (SMD) together with the 95% confidence intervals (95% CI). Statistical significance was set at $P \leq 0.05$. The scale used for interpretation of the magnitude of the effect size was specific to training research: negligible (<0.2), small ($0.2-0.5$), moderate ($0.5-0.8$), and large (≥ 0.8).⁵⁹ Consistency (i.e. homogeneity) of effects for each dependent variable was assessed using the Cochran's Q and I^2 .⁶⁰

Results

Description of included studies

A total of 117 effects were analysed from 44 original studies. The combined method was the conditioning programme most utilised (44 effects; 1121 participants: 539 control and 582 experimental), followed by plyometric training (34 effects; 920 participants: 453 control and 467 experimental), sprint or change of direction training (24 effects; 526 participants: 259 control and 267 experimental), and resistance training (15 effects; 345 participants: 174 control and 171 experimental).

Collectively, exercise intervention length ranged from 6 to 26 weeks (8.8 ± 3.5 weeks) and the number of conditioning training sessions per week ranged from 1 to 5 (2.1 ± 0.7 weeks). Twenty-nine interventions were conducted during the in-season, 12 during the pre-season, and four did not report the phase of the season. Thirty-six interventions were conducted by men, 8 by women, and 1 included both sexes. The age of the participants ranged from 14 to 25 years. The general characteristics of the studies included in the systematic review and meta-analysis are summarised in Table 1.

The quality of the 44 studies included in the meta-analysis is summarised in Table 2. The median PeDro score was 5 of 9. All studies clearly stated eligibility criteria and provided point estimates for effect size calculation. Twenty-seven (61%) studies were randomised and 43 (98%) studies matched intervention groups at baseline. Only two (5%) studies used evaluators blinded to treatment allocation. Seventeen (39%) studies reported that $>85\%$ of participants had complied with the intervention. Thirteen (30%) studies awarded a point in the intention-to-treat analysis, but in all cases these studies reported that all participants received treatment or control conditions as allocated. Forty (91%) studies completed between-group analyses. Only one (2%) study used concealed allocation.

Vertical jump ability

Thirty-four effects were analysed from 29 original studies (Figure 2). The performance of conditioning programmes in addition to regular soccer training was associated with a significant increase in vertical jump performance (SMD = 0.64 [95% CI: 0.50 to 0.77], $P < 0.00001$). Concerning the subgroup analyses, significant performance improvements were observed after resistance training (SMD = 0.57 [95% CI: 0.14 to 1.01], $P = 0.01$), plyometric training (SMD = 0.83 [95% CI: 0.53–1.13], $P < 0.00001$), sprint or change of direction training (SMD = 0.70 [95% CI: 0.30–1.09], $P = 0.0006$), and combined methods (SMD = 0.49 [95% CI: 0.27–0.71], $P < 0.0001$). No significant differences among subgroups were observed ($P = 0.34$).

Four additional studies measured at least one outcome related to the vertical jump ability; however, they were excluded from the meta-analysis because they failed to report the mean and SD values of the outcomes measured.^{5,18,46,34} Among these studies, three reported net treatment effect significantly in favour of the experimental group,^{18,46,34} one reported no significant differences between the control and experimental group,⁵ and no study reported net treatment effect significantly in favour of the control group.

Linear sprint performance

Forty-two effects were analysed from 33 original studies (Figure 3). The performance of conditioning programmes in addition to regular soccer training was associated with a significant reduction in linear sprint time (SMD = -0.79 [95% CI: -0.61 to -0.96], $P < 0.00001$). Concerning the subgroup analyses, significant performance improvements were observed after resistance training (SMD = -0.76 [95% CI: -0.37 to -1.15], $P = 0.0001$), plyometric training

Table 2. Quality metrics of included studies.

Study name	Eligibility criteria specified	Random allocation of participants	Allocation concealed	Groups similar at baseline	Assessors blinded	Outcome measures assessed in 85% of participants	Intention to treat analysis	Reporting of between group statistical comparison	Point measures and measures of variability reported	Overall PeDro score
Brito et al. ⁵	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
Cavaco et al. ⁴¹	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Chaouachi et al. ³²	Yes	Yes	No	yes	No	No	No	Yes	Yes	5
Chelly et al. ⁶	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Chelly et al. ¹⁷	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
De Hoyo et al. ⁷	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Dupont et al. ³³	Yes	No	No	Yes	No	No	No	No	Yes	3
Enoksen et al. ⁸	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
Fábrica et al. ¹⁹	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Faude et al. ⁴²	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Franco-Márquez et al. ⁴³	Yes	No	No	Yes	No	Yes	No	Yes	Yes	5
García-Pinillos et al. ⁴⁴	Yes	Yes	No	Yes	No	Yes	No	Yes	Yes	6
Gauffin et al. ²⁰	Yes	No	No	Yes	No	Yes	No	Yes	Yes	5
González-Badillo et al. ⁴⁵	Yes	No	No	No	No	No	No	Yes	Yes	3
Haghighi et al. ⁹	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Hammami et al. ²¹	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
Hernández et al. ⁴⁷	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Hernández et al. ⁴⁸	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Jelusic et al. ¹⁰	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Karsten et al. ¹²	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
Kavaliuskas et al. ³⁵	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Kumar ²²	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Kumar ²³	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
López-Segovia et al. ⁴⁹	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Maio Alves et al. ⁵⁰	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Manolopoulos et al. ¹³	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
Mathisen et al. ³⁶	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Nedrehagen et al. ³⁷	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Ozbar ²⁵	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
Ozbar et al. ²⁶	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
Polman et al. ³⁸	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Ramirez-Campillo et al. ²⁸	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	8
Ramirez-Campillo et al. ²⁸	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	8
Rønnestad et al. ¹⁵	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Sáez de Villarreal et al. ⁵²	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Sanchez-Sanchez et al. ¹⁶	Yes	No	No	Yes	No	No	No	No	Yes	3
Sedano et al. ²⁹	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	7
Sedano et al. ³⁰	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Shalfawi et al. ³⁹	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Siegler et al. ⁵³	Yes	No	No	Yes	No	No	No	No	Yes	3
Singh et al. ⁵⁴	Yes	No	No	Yes	No	No	No	Yes	Yes	4
Tønnessen et al. ⁴⁰	Yes	Yes	No	Yes	No	No	No	Yes	Yes	5
Vaczi et al. ³¹	Yes	Yes	No	Yes	No	Yes	Yes	No	Yes	6
Wong et al. ⁵⁵	Yes	No	No	Yes	No	Yes	Yes	Yes	Yes	6

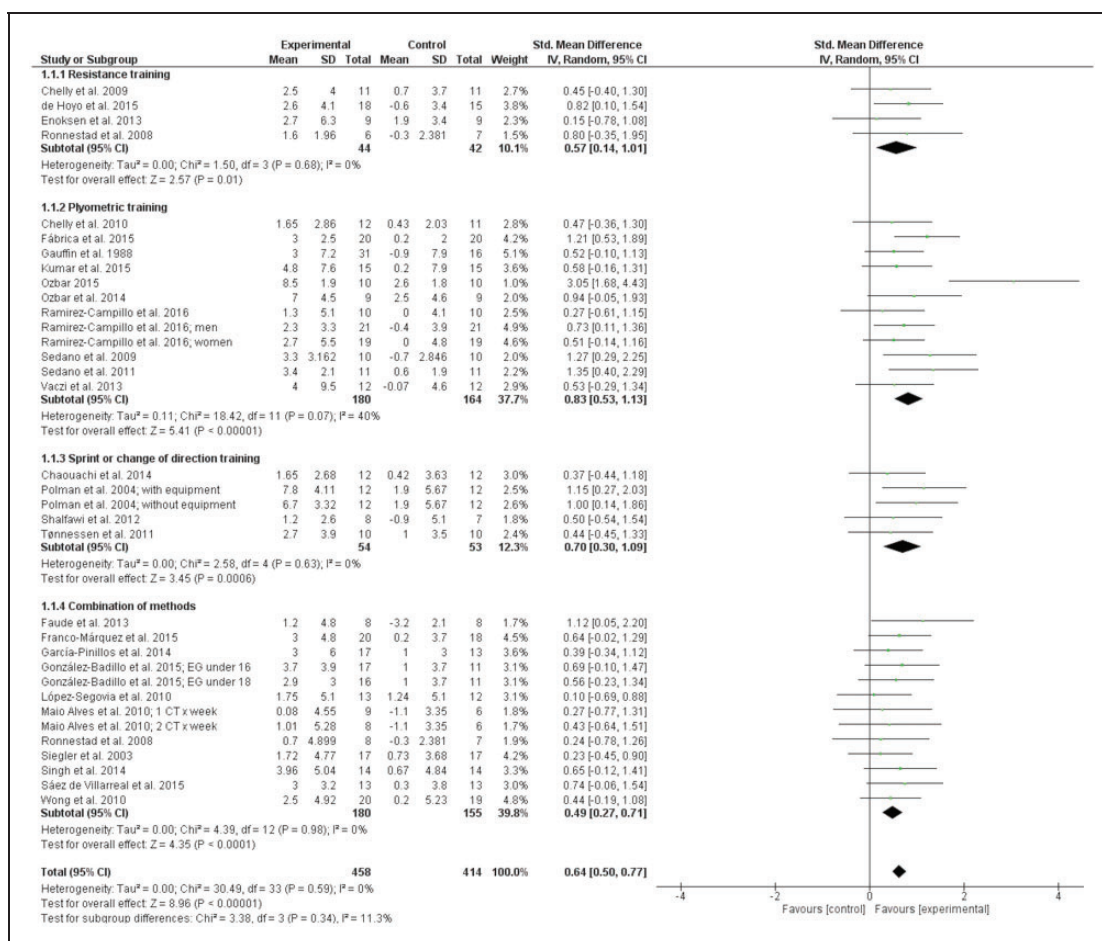


Figure 2. Effects of conditioning programmes applied in addition to regular soccer training as compared to soccer training alone on vertical jump performance. Forest plot shows pooled standardised mean differences with 95% confidence intervals (CI) for 29 controlled trials. Subgroup analyses show the results for each type of conditioning programme: resistance training (4), plyometric training (12), sprint and/or change of direction training (5) and combined methods (13). The diamond at the bottom of the graph and the subgroups represents the pooled standardised mean difference with the 95% CI for all trials following random effect meta-analyses.

($SMD = -1.09$ [95% CI: -0.82 to -1.36], $P < 0.00001$), sprint or change of direction training ($SMD = -0.82$ [95% CI: -0.39 to -1.24], $P = 0.0002$), and combined methods ($SMD = -0.62$ [95% CI: -0.32 to -0.96], $P < 0.0001$). No significant differences among subgroups were observed ($P = 0.13$).

Seven additional studies measured at least one outcome related to the linear sprint ability; however, they were excluded from the meta-analysis because they failed to report the mean and SD values of the outcomes measured.^{18,46,34,11,51,13,24} Among these studies, four reported net treatment effect significantly in favour of the experimental group,^{18,34,51,24} four reported no significant differences between the control and experimental group,^{18,46,11,13} and no study reported net treatment effect significantly in favour of the control group.

Change of direction with ball

Seven effects were analysed from five original studies (Figure 4). The performance of conditioning programmes in addition to regular soccer training was associated with a significant increase in change of direction with ball ability ($SMD = -0.78$ [95% CI: -0.42 to -1.15], $P < 0.0001$). Concerning the subgroup analyses, significant performance improvements were observed after plyometric training ($SMD = -1.48$ [95% CI: -0.47 to -2.5], $P = 0.004$) and combined methods ($SMD = -0.81$ [95% CI: -0.29 to -1.33], $P = 0.002$). No significant differences were observed for resistance training ($SMD = -0.87$ [95% CI: 0.06 to -1.79], $P = 0.07$), and sprint or change of direction training ($SMD = -0.22$ [95% CI: 0.58 to -1.03], $P = 0.58$). No significant differences among subgroups were observed ($P = 0.29$).

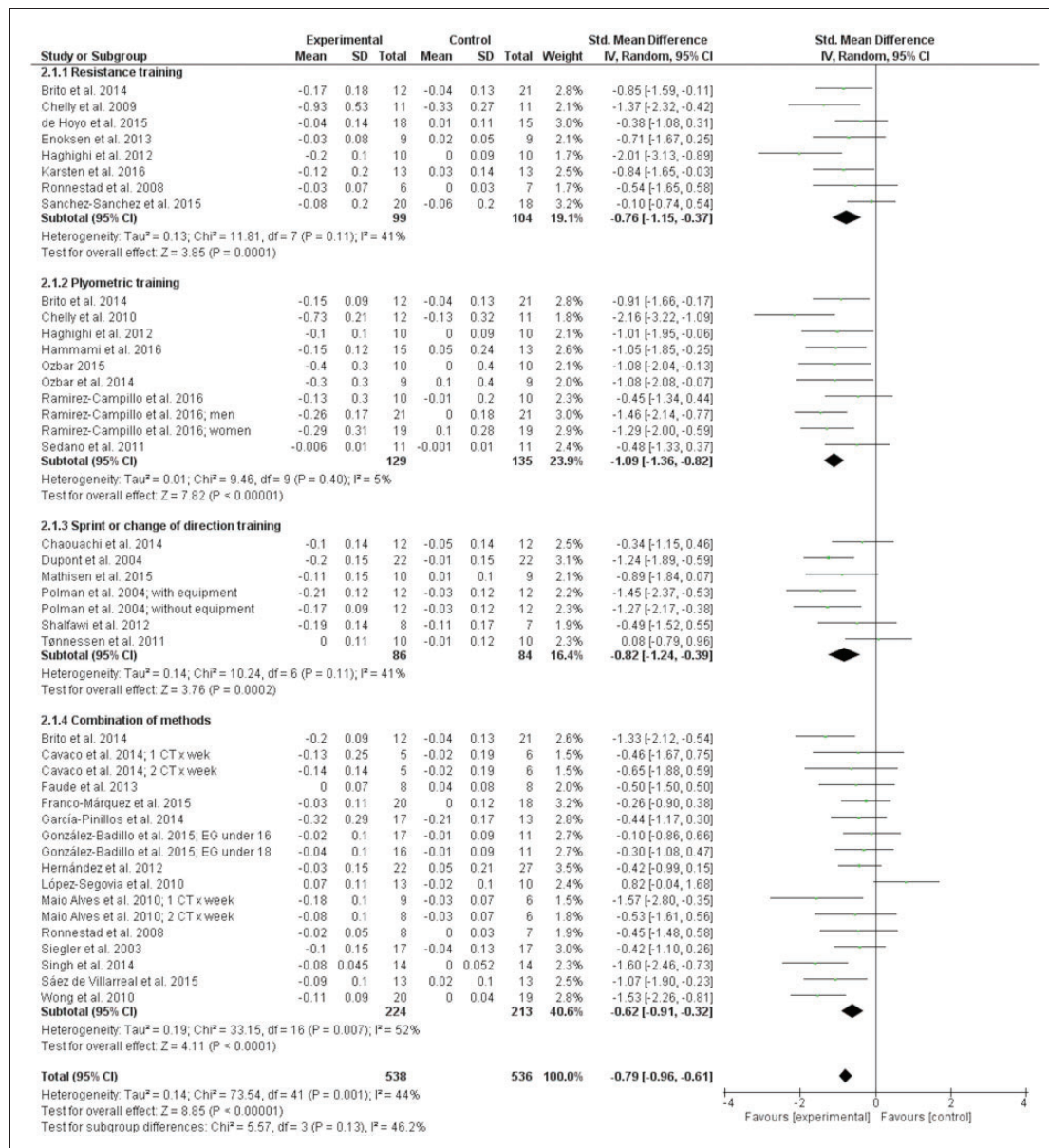


Figure 3. Effects of conditioning programmes applied in addition to regular soccer training as compared to soccer training alone on linear sprint performance. Forest plot shows pooled standardised mean differences with 95% confidence intervals (CI) for 33 controlled trials. Subgroup analyses show the results for each type of conditioning programme: resistance training (8), plyometric training (10), sprint and/or change of direction training (7) and combined methods (17). The diamond at the bottom of the graph and the subgroups represents the pooled standardised mean difference with the 95% CI for all trials following random effect meta-analyses.

Two additional studies measured at least one outcome related to the change of direction with ball ability; however, they were excluded from the meta-analysis because they failed to report the mean and SD values of the outcomes measured.^{11,24} Among these studies, one reported net treatment effect significantly in favour of the experimental group,²⁴ one reported no significant differences between the control and experimental group,¹¹ and no study reported net

treatment effect significantly in favour of the control group.

Change of direction without ball

Eighteen effects were analysed from 15 original studies (Figure 5). The performance of conditioning programmes in addition to regular soccer training was associated with a significant increase in change of

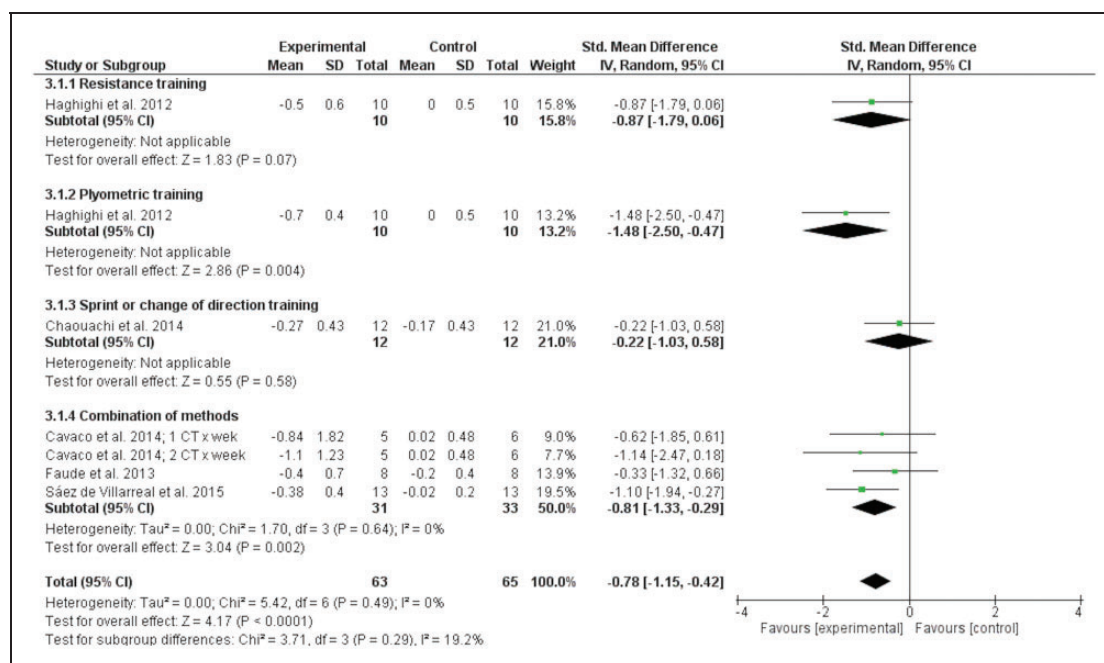


Figure 4. Effects of conditioning programmes applied in addition to regular soccer training as compared to soccer training alone on change of direction with ball performance. Forest plot shows pooled standardised mean differences with 95% confidence intervals (CI) for five controlled trials. Subgroup analyses show the results for each type of conditioning programme: resistance training (1), plyometric training (1), sprint and/or change of direction training (1) and combined methods (4). The diamond at the bottom of the graph and the subgroups represents the pooled standardised mean difference with the 95% CI for all trials following random effect meta-analyses.

direction without ball ability (SMD = -0.83 [95% CI: -0.61 to -1.05], $P < 0.00001$). Concerning the subgroup analyses, significant performance improvements were observed after plyometric training (SMD = -0.87 [95% CI: -0.44 to -1.31], $P < 0.0001$), sprint or change of direction training (SMD = -0.82 [95% CI: -0.39 to -1.24], $P = 0.0002$), and combined methods (SMD = -0.81 [95% CI: -0.44 to -1.17], $P < 0.0001$). No significant differences among subgroups were observed ($P = 0.97$).

Two additional studies measured at least one outcome related to the change of direction without ball ability; however, they were excluded from the meta-analysis because they failed to report the mean and SD values of the outcomes measured.^{5,24} Among these studies, one reported net treatment effect significantly in favour of the experimental group,²⁴ one reported no significant differences between the control and experimental group,⁵ and no study reported net treatment effect significantly in favour of the control group.

Repeated sprint ability

Seven effects were analysed from six original studies (Figure 6). The performance of conditioning programmes in addition to regular soccer training was

associated with a significant increase in repeated sprint performance (SMD = -0.51 [95% CI: -0.19 to -0.84], $P = 0.002$). Concerning the subgroup analyses, significant performance improvements were observed after sprint or change of direction training (SMD = -0.52 [95% CI: -0.13 to -0.91], $P = 0.009$). However, although obtaining comparable effect sizes, the improvements observed after plyometric training (SMD = -0.50 [95% CI: 0.08 to -1.07], $P = 0.09$) did not reach statistical significance. No significant differences between subgroups were observed ($P = 0.94$).

One additional study measured at least one outcome related to the linear sprint ability; however, they were excluded from the meta-analysis because they failed to report the mean and SD values of the outcomes measured.¹¹ This study reported no significant differences between the control and experimental group.¹¹

Kicking performance

Nine effects were analysed from nine original studies (Figure 7). The performance of conditioning programmes in addition to regular soccer training was associated with a significant increase in kicking performance (SMD = 0.78 [95% CI: 0.34 – 1.22], $P = 0.0005$). Concerning the subgroup analyses, significant performance improvements were observed after

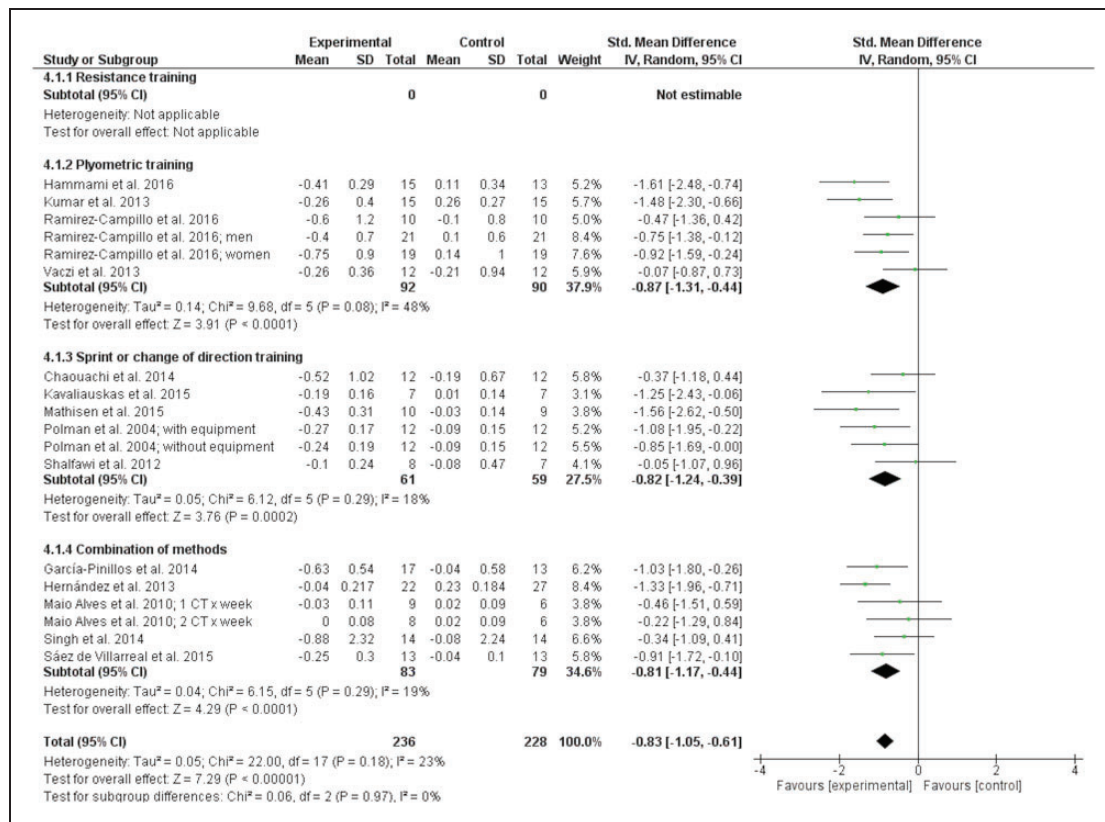


Figure 5. Effects of conditioning programmes applied in addition to regular soccer training as compared to soccer training alone on change of direction without ball performance. Forest plot shows pooled standardised mean differences with 95% confidence intervals (CI) for 15 controlled trials. Subgroup analyses show the results for each type of conditioning programme: resistance training (0), plyometric training (6), sprint and/or change of direction training (6) and combined methods (6). The diamond at the bottom of the graph and the subgroups represents the pooled standardised mean difference with the 95% CI for all trials following random effect meta-analyses.

resistance training ($SMD = 0.76$ [95% CI: 0.08–1.45], $P = 0.03$) and plyometric training ($SMD = 1.43$ [95% CI: 0.78–2.09], $P < 0.0001$). No significant differences were observed for combined methods ($SMD = 0.40$ [95% CI: -0.13 to 0.94], $P = 0.14$). No significant differences among subgroups were observed ($P = 0.06$).

Two additional studies measured at least one outcome related to the kicking performance; however, they were excluded from the meta-analysis because they failed to report the mean and SD values of the outcomes measured.^{14,51} Both studies reported net treatment effect significantly in favour of the experimental group.^{14,51}

Discussion

The purpose of this meta-analysis was to determine whether, and at which extent different conditioning programmes implemented in addition to regular soccer training are able to enhance the performance of high-velocity soccer-related tasks. The obtained data support our hypothesis since significant

improvements in the experimental groups (soccer training + conditioning programme) compared to the control groups (soccer training alone) were observed in all six outcomes analysed. The magnitude of the improvements was moderate for five outcomes: vertical jump, linear sprint, change of direction with ball, repeated sprint ability and kicking performance; and large for the change of direction without ball category.

Vertical jump ability

Significant improvements in vertical jump performance were obtained from the four types of conditioning programmes. The magnitude of the improvements was deemed small for the combination of methods ($SMD = 0.49$), moderate for resistance training ($SMD = 0.57$) and sprint or change of direction training ($SMD = 0.70$), and large for the plyometric training ($SMD = 0.83$). However, the differences observed among the groups were not significant. Therefore, the present meta-analysis cannot provide conclusive information regarding the best conditioning programme

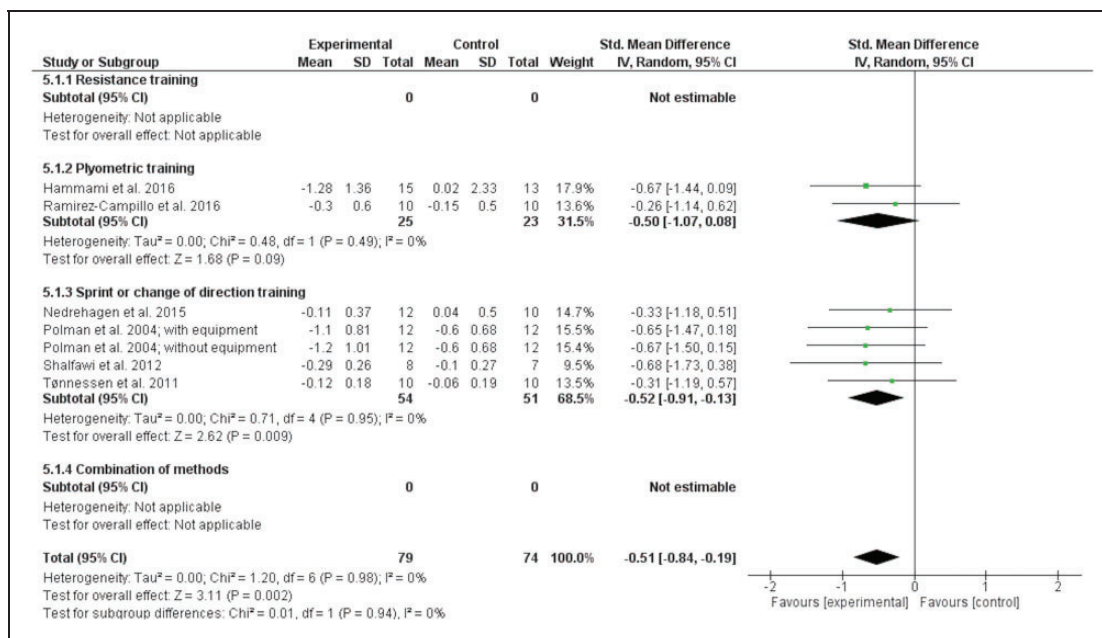


Figure 6. Effects of conditioning programmes applied in addition to regular soccer training as compared to soccer training alone on repeated sprint ability performance. Forest plot shows pooled standardised mean differences with 95% confidence intervals (CI) for six controlled trials. Subgroup analyses show the results for each type of conditioning programme: resistance training (0), plyometric training (2), sprint and/or change of direction training (5) and combined methods (0). The diamond at the bottom of the graph and the subgroups represents the pooled standardised mean difference with the 95% CI for all trials following random effect meta-analyses.

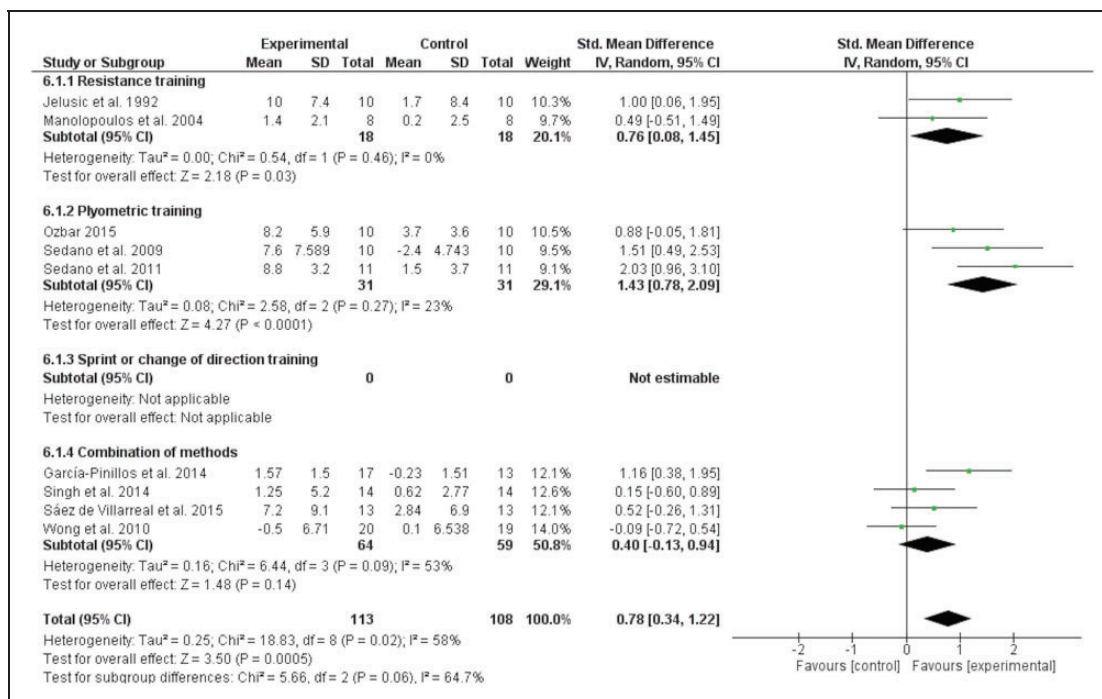


Figure 7. Effects of conditioning programmes applied in addition to regular soccer training as compared to soccer training alone on kicking performance. Forest plot shows pooled standardised mean differences with 95% confidence intervals (CI) for nine controlled trials. Subgroup analyses show the results for each type of conditioning programme: resistance training (2), plyometric training (3), sprint and/or change of direction training (0), and combined methods (4). The diamond at the bottom of the graph and the subgroups represents the pooled standardised mean difference with the 95% CI for all trials following random effect meta-analyses.

aimed to increase vertical jump performance in soccer players.

Plyometric training could be a good choice to increase unloaded vertical jump performance since this training method potentiates lower-body muscular function without inducing significant muscle hypertrophy.⁶¹ On the other hand, resistance training, especially if it is oriented towards muscle hypertrophy (moderate loads [6-12 RM] mobilised at slow velocities) as was the case in several studies included in this meta-analysis,^{5,8} could be counterproductive to this goal as it can result in cumulative fatigue and increases in body mass which collectively might compromise jump performance. In addition, one could expect greater improvements in jump performance after plyometric training than sprint or change of direction training because the direction of force application is more specific in plyometric training when vertical jumps are performed.⁶² Finally, the non-significant Chi^2 test ($P=0.59$) together with the $I^2=0\%$ reflects the consistency of the effect sizes included in the present meta-analysis.

Linear sprint performance

The present meta-analysis indicates that linear sprint performance can be enhanced after a short-term conditioning programme applied concurrently with soccer training. The improvements were significant for the four types of conditioning programmes. The magnitude of the improvements was deemed moderate for the combination of methods ($\text{SMD}=-0.62$) and resistance training ($\text{SMD}=-0.76$), and large for the sprint or change of direction training ($\text{SMD}=-0.82$) and plyometric training ($\text{SMD}=-1.09$). However, the subgroup differences were not significant, and consequently the present meta-analysis cannot provide conclusive information regarding the best conditioning programme to increase linear sprint performance in soccer players.

In line with the results obtained for the vertical jump category, it seems that short-term plyometric and sprint training may be more effective at enhancing linear sprint performance than resistance training alone. One plausible explanation of these results might be that in many of the resistance training programmes included in the meta-analysis, the participants were not instructed to perform each repetition at the maximal intended velocity. This characteristic could have reduced the potential benefits of these training interventions as it is known that performing each repetition at the maximum possible speed induces greater adaptations.⁶³ Therefore, future resistance training studies should ensure that participants perform all repetitions at the maximum possible speed as happens in the plyometric and sprint training methods. Another explanation could be that the type of resistance training

generally used does not maximise strength gains, and therefore it could not transfer positively to sprint performance as noted by Seitz et al.⁶⁴ Finally, the significant Chi^2 test ($P=0.001$) together with the $I^2=44\%$ reflects a moderate degree of inconsistency in the effect sizes included in the present meta-analysis.⁶⁰

Change of direction with ball

The pooled effects of the four conditioning programmes reveal that change of direction with ball performance can be enhanced after a short-term conditioning programme applied concurrently with regular soccer training. The improvement only reached statistical significance for the combination of methods category. However, the small sample size of the three remaining training categories (10-12 participants per training group) does not allow for firm conclusions to be made because of a low precision. In this regard, the magnitudes of the improvements were large for resistance training ($\text{SMD}=-0.87$), plyometric training ($\text{SMD}=-1.48$), and the combination of methods ($\text{SMD}=-0.81$). The magnitude of the improvement was small ($\text{SMD}=-0.22$) for the only study that evaluated the change of direction ability with a soccer ball after sprint training.³² The test for subgroup differences did not reveal significant differences. In addition, the non-significant Chi^2 test ($P=0.49$) together with the $I^2=0\%$ reflects the consistency of the effect sizes included in the present meta-analysis from the different conditioning programmes.

Besides physical ability, the performance of change of direction with a soccer ball could also be affected by soccer players' technical ability. The potential motor learning effects could be considered to be a potential confounding factor when testing the ability to change direction, especially if the test is conducted with a ball. However, motor learning should have equally influenced both the control and experimental groups included in the present meta-analysis, because of the level of soccer players included in the selected studies. Therefore, the benefits observed in the change of direction with the ball category for the experimental group compared to the control group seem to be attributed to the enhancement in the soccer players' physical ability. In this regard, the results of the present meta-analysis suggest that similar improvements in agility with and without the ball could be expected after a short-term conditioning programme ($\text{SMD}=-0.78$ and -0.83 , respectively).

Change of direction without ball

The present meta-analysis indicates that change of direction without the ball performance can be enhanced

after a short-term conditioning programme applied concurrently with soccer specific training. The improvements were significant and large for the three types of conditioning programmes that monitored this ability: plyometric training ($SMD = -0.87$), sprint or change of direction training ($SMD = -0.82$), and training which included a combination of methods ($SMD = -0.81$). The subgroup differences were not significant, and consequently the present meta-analysis cannot provide conclusive information regarding the best conditioning programme to increase change of direction without ball performance in soccer players. The non-significant χ^2 test ($P = 0.18$) together with the low I^2 (23%) reflect the consistency of the effect sizes included in the present meta-analysis.

Repeated sprint ability

The improvements in repeated sprint ability were significant but moderate for the two types of conditioning programmes that monitored this ability: plyometric training ($SMD = -0.50$) and sprint or change of direction training ($SMD = -0.52$). The non-significant χ^2 test ($P = 0.98$) together with the $I^2 = 0\%$ reflects the consistency of the effect sizes included in the present meta-analysis.

The results of this meta-analysis suggest that the expected improvement of repeated sprint ability ($SMD = -0.51$) could be lower than for the other five outcomes examined (SMD from 0.64 to 0.83). The lack of aerobic training stimulus in the conditioning programmes followed by the experimental groups may not be responsible for this result because repeated sprint ability seems to be poorly correlated with maximal oxygen uptake.⁶⁵ Repeated sprint ability is a complex fitness component that depends on both neural and metabolic factors.⁶⁶ The plyometric and sprint conditioning programmes included in this meta-analysis have mainly focused on enhancing the neural factors, which are expected to improve single-sprint performance. Therefore, we speculate that higher increments in repeated sprint ability would have been reached if the training programmes would have included high-intensity interval training that is expected to improve the ability to recover between sprints.^{66,67}

The comparison of the subgroups did not reveal significant differences between the plyometric and sprint or change of direction training protocols ($P = 0.94$). However, a study conducted by Buchheit et al.⁶² shows that the repeated shuttle sprint training could be more effective than explosive strength training to improve repeated sprint ability in young elite soccer players. Curiously, Buchheit et al.⁶² also showed that explosive strength training which mainly consisted on plyometric exercises was a more effective method for

improving vertical jump ability. These results support the importance of training specificity.

Kicking performance

The present meta-analysis indicates that kicking performance in terms of ball speed can be enhanced after a short-term conditioning programme applied concurrently with soccer specific training methods. The improvements were significant and large for plyometric training ($P < 0.0001$; $SMD = 1.43$), significant and moderate for resistance training ($P = 0.03$; $SMD = 0.76$), and non-significant and small for the combination of methods ($P = 0.03$; $SMD = 0.76$). The test for subgroup differences was close to be statically significant ($P = 0.06$). The significant χ^2 test ($P = 0.02$) together with the $I^2 = 58\%$ reflects a moderate degree of inconsistency in the effect sizes included in the present meta-analysis.⁶⁰

Only two studies reported trivial net treatment effect size for kicking performance.^{54,55} Both studies belong to the combined method category. As noted by Wong et al.,⁵⁵ the nature of the resistance training exercises performed in these studies, which was not focus on the development of movement speed, could partially explain these results. The greater similarity of the plyometric exercises, which require powerful stretch-shortening muscle actions of hip and leg muscles as happens in the soccer kick, might explain their superior results. However, further studies are needed to directly compare the effect of different conditioning programmes in the short and long terms on kicking performance in soccer players.

Limitations and future research

Besides the inherent limitations associated with the meta-analytic technique itself, a number of specific limitations of the current meta-analysis have to be considered. Regarding the primary literature, we should acknowledge that several important scientific criteria such as evaluators blinded to treatment allocation (5%) or concealed allocation (2%) were rarely met. This meta-analysis does not allow us to provide definite conclusions regarding which is the most effective conditioning programme to enhance each of the outcomes analysed. However, the present meta-analysis indicates that different conditioning programmes implemented in addition to regular soccer training enhance the performance of high-velocity soccer-related tasks. Therefore, based on the positive effects described in the current scientific literature, further studies comparing the effect of different conditioning programmes against a control group that only perform soccer training are not needed. In this regard, we encourage future

studies to compare the effects of different conditioning programmes^{62,68} or different variants within the same type of conditioning programme^{69,70} to elucidate which are the most beneficial conditioning programmes. Based on the recent study conducted by Ramírez-Campillo et al.²⁸ who did not find an interaction for sex after six weeks of plyometric training, we decided to pool the data of men and women in the meta-analysis to include a broader number of studies. However, the specific information of each study presented in Table 1 together with the absolute values provided in Figures 2 to 7 for both control and experimental groups allows the interested readers to re-conduct the meta-analysis if they want to delimitate the range of studies even more.

Conclusions

The implementation of a conditioning programme in addition to regular soccer specific training methods significantly enhances the performance of high-velocity soccer-related tasks in soccer players. The magnitude of the improvements was moderate for vertical jump (SMD = 0.64), linear sprint (SMD = -0.79), change of direction with ball (SMD = -0.78), repeated sprint ability (SMD = -0.51), and kicking performance (SMD = 0.78); and large for the change of direction without ball category (SMD = -0.83). These results suggest that coaches and physical trainers should implement conditioning programmes in conjunction with traditional soccer training in order to enhance the performance of several soccer specific skills that might be determinant of success during soccer competitions. Although it is not possible to provide firm conclusions from the results of this meta-analysis, short-term plyometric training seems to be an ideal method to potentiate lower-body muscular function. In this regard, plyometric training was the conditioning programme that provided the largest effect size in five of the six outcomes examined.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

References

1. Kunz M. Big count: 265 million playing football. *FIFA Mag* 2007; 10–15. Available at: http://www.fifa.com/mm/document/fifafacts/bcoffsurv/emaga_9384_10704.pdf.
2. Stølen T, Chamari K, Castagna C, et al. Physiology of soccer: an update. *Sports Med* 2005; 35: 501–536.
3. Faude O, Koch T and Meyer T. Straight sprinting is the most frequent action in goal situations in professional football. *J Sports Sci* 2012; 30: 625–631.
4. Bishop DJ and Girard O. Determinants of team-sport performance: Implications for altitude training by team-sport athletes. *Br J Sports Med* 2013; 47: i17–i21.
5. Brito J, Vasconcellos F, Oliveira JJ, et al. Short-term performance effects of three different low-volume strength-training programmes in college male soccer players. *J Hum Kinet* 2014; 40: 121–128.
6. Chelly MS, Fathloun M, Cherif N, et al. Effects of a back squat training program on leg power, jump, and sprint performances in junior soccer players. *J Strength Cond Res* 2009; 23: 2241–2249.
7. de Hoyo M, Pozzo M, Sanudo B, et al. Effects of a 10-week in-season eccentric overload training program on muscle injury prevention and performance in junior elite soccer players. *Int J Sports Physiol Perform* 2015; 10: 46–52.
8. Enoksen E, Staxrud M, Tonnessen E, et al. The effect of supervised strength training on young elite male soccer players' physical performance. *Serb J Sports Sci* 2013; 7: 173–179.
9. Haghighi A, Moghadasi M, Nikseresht A, et al. Effects of plyometric versus resistance training on sprint and skill performance in young soccer players. *Eur J Exp Biol* 2012; 2: 2348–2351.
10. Jelusic V, Jaric S and Kukolj M. Effects of the stretch-shortening strength training on kicking performance in soccer players. *J Hum Mov Stud* 1992; 22: 231–238.
11. Jullien H, Bisch C, Largouët N, et al. Does a short period of lower limb strength training improve performance in field-based tests of running and agility in young professional soccer players? *J Strength Cond Res* 2008; 22: 404–411.
12. Karsten B, Larumb-Zabala E, Kandemir G, et al. The effects of a 6-week strength training on critical velocity, anaerobic running distance, 30-M sprint and Yo-Yo intermittent running test performances in male soccer players. *PLoS One* 2016; 11: 1–10.
13. Manolopoulos E, Papadopoulos C, Salonikidis K, et al. Strength training effects on physical conditioning and instep kick kinematics in young amateur soccer players during preseason. *Percept Mot Ski* 2004; 99: 701–710.
14. Manolopoulos E, Katis A, Manolopoulos K, et al. Effects of a 10-week resistance exercise program on soccer kick biomechanics and muscle strength. *J Strength Cond Res* 2013; 27: 3391–3401.
15. Ronnestad BR, Kvamme NH, Sunde A, et al. Short-term effects of strength and plyometric training on sprint and jump performance in professional soccer players. *J Strength Cond Res* 2008; 22: 773–780.
16. Sánchez-Sánchez J, Pérez S, Yagüe JM, et al. Implementation of a resistance training on young football players. *Rev Int Med Cienc Act Fis Deporte* 2015; 15: 45–59.
17. Chelly M, Ghenem M, Abid K, et al. Effects of in-season short-term plyometric training program on leg power, jump- and sprint performance of soccer players. *J Strength Cond Res* 2010; 24: 2670–2676.

18. Chettiamkudiyil HB, Meethal A and Najeeb AM. Effect of plyometric training and saq training on explosive strength and speed of university level soccer players. *Int J Heal Phys Educ Comput Sci Sport* 2015; 17: 170–173.
19. Fábrega G, López F and Souto A. Effects of power training in mechanical stiffness of the lower limbs in soccer players. *Rev Andal Med Deporte* 2015; 8: 145–149.
20. Gauffin H, Ekstrand J and Tropp H. Improvement of vertical jump performance in soccer players after specific training. *J Hum Mov Stud* 1988; 15: 185–190.
21. Hammami M, Negra Y, Aouadi R, et al. Effects of an in-season plyometric training program on repeated change of direction and sprint performance in the junior soccer player. *J Strength Cond Res* 2016; 30: 3312–3320.
22. Kumar R. The effect of 6 week plyometric training program on agility of collegiate soccer players. *Int J Behav Soc Mov Sci* 2013; 2: 170–176.
23. Kumar R. The effect of 6 week plyometric training program on maximal vertical jumping height of collegiate level soccer players. *Int J Appl Res* 2015; 1: 385–389.
24. Memarzadeh A, Moghadasi M and Zare K. Effects of plyometric training on skill performance in soccer players. *Int J Curr Res Aca Rev* 2014; 2: 242–247.
25. Ozbar N. Effects of plyometric training on explosive strength, speed and kicking speed in female soccer players. *Anthropologist* 2015; 19: 333–339.
26. Ozbar N, Ates S and Agopya A. The effect of 8-week plyometric training on leg power, jump and sprint performance in female soccer players. *J Strength Cond Res* 2014; 28: 2888–2894.
27. Ramírez-Campillo R, González-Jurado JA, Martínez C, et al. Effects of plyometric training and creatine supplementation on maximal-intensity exercise and endurance in female soccer players. *J Sci Med Sport* 2016; 19: 682–687.
28. Ramírez-Campillo R, Vergara-Pedrerros M, Henríquez-Olguín C, et al. Effects of plyometric training on maximal-intensity exercise and endurance in male and female soccer players. *J Sports Sci* 2016; 34: 687–693.
29. Sedano Campo S, Vaeyens R, Philippaerts R, et al. Effects of lower-limb plyometric training on body composition, explosive strength, and kicking speed in female soccer players. *J Strength Cond Res* 2009; 23: 1714–1722.
30. Sedano S, Matheu A, Redondo JC, et al. Effects of plyometric training on explosive strength, acceleration capacity and kicking speed in young elite soccer players. *J Sports Med Phys Fitness* 2011; 51: 50–58.
31. Vácz M, Tollár J, Meszler B, et al. Short-term high intensity plyometric training program improves strength, power and agility in male soccer players. *J Hum Kinet* 2013; 37: 17–26.
32. Chaouachi A, Chtara M, Hammami R, et al. Multidirectional sprints and small-sided games training effect on agility and change of direction abilities in youth soccer. *J Strength Cond Res* 2014; 28: 3121–3127.
33. Dupont G, Akakpo K and Berthoin S. The effect of in-season, high-intensity interval training in soccer players. *J Strength Cond Res* 2004; 18: 584–589.
34. Jovanovic M, Sporis G, Omrcen D, et al. Effects of speed, agility, quickness training method on power performance in elite soccer players. *J Strength Cond Res* 2011; 25: 1285–1292.
35. Kavaliauskas M, Kilvington R and Babraj J. Effects of in-season uphill sprinting on physical characteristics in semi-professional soccer players. *J Sports Med Phys Fitness* 2015; 57: 165–170.
36. Mathisen GE and Pettersen SA. The effect of speed training on sprint and agility performance in 15-year-old female soccer players. *LASE J Sport Sci* 2015; 6: 63–72.
37. Nedrehagen ES and Saeterbakken AH. The effects of in-season repeated sprint training compared to regular soccer training. *J Hum Kinet* 2015; 49: 237–244.
38. Polman R, Walsh D, Bloomfield J, et al. Effective conditioning of female soccer players. *J Sports Sci* 2004; 22: 191–203.
39. Shalfawi SAI, Ingebrigtsen J, Dillern T, et al. The effect of 40 m repeated sprint training on physical performance in young elite male soccer players. *Serb J Sports Sci* 2012; 6: 111–116.
40. Tonnessen E, Shalfawi S, Haugen T, et al. The effect of 40-m repeated sprint training on maximum sprinting speed, repeated sprint speed endurance, vertical jump, and aerobic capacity in young elite male soccer players. *J Strength Cond Res* 2011; 25: 2364–2370.
41. Cavaco B, Sousa N, Dos Reis VM, et al. Short-term effects of complex training on agility with the ball, speed, efficiency of crossing and shooting in youth soccer players. *J Hum Kinet* 2014; 43: 105–112.
42. Faude O, Roth R, Di Giovine D, et al. Combined strength and power training in high-level amateur football during the competitive season: a randomised-controlled trial. *J Sports Sci* 2013; 31: 1460–1467.
43. Franco-Márquez F, Rodríguez-Rosell D, González-Suárez JM, et al. Effects of combined resistance training and plyometrics on physical performance in young soccer players. *Int J Sports Med* 2015; 36: 906–914.
44. García-Pinillos F, Martínez-Amat A, Hita-Contreras F, et al. Effects of a contrast training program without external load on vertical jump, kicking speed, sprint, and agility of young soccer players. *J Strength Cond Res* 2014; 28: 2452–2460.
45. González-Badillo J, Pareja-Blanco F, Rodríguez-Rosell D, et al. Effects of velocity-based resistance training on young soccer players of different ages. *J Strength Cond Res* 2015; 29: 1329–1338.
46. Gorostiaga EM, Izquierdo M, Ruesta M, et al. Strength training effects on physical performance and serum hormones in young soccer players. *Eur J Appl Physiol* 2004; 91: 698–707.
47. Hernández YH and García JM. Efectos de un entrenamiento específico de potencia aplicado a futbolistas juveniles para la mejora de la velocidad lineal. *Mot Eur J Hum Mov* 2012; 28: 125–144.
48. Hernández YH and García JM. Efectos de un entrenamiento específico de potencia aplicado a futbolistas juveniles para la mejora de la velocidad con cambio de dirección. *Mot Eur J Hum Mov* 2013; 31: 17–36.
49. López-Segovia M, Palao Andrés JM and González-Badillo JJ. Effect of 4 months of training on aerobic

- power, strength, and acceleration in two under-19 soccer teams. *J Strength Cond Res* 2010; 24: 2705–2714.
50. Maio Alves J, Rebelo A, Abrantes C, et al. Short-term effects of complex and contrast training in soccer players' vertical jump, sprint, and agility abilities. *J Strength Cond Res* 2010; 24: 936–941.
 51. Manolopoulos E, Papadopoulos C and Kellis E. Effects of combined strength and kick coordination training on soccer kick biomechanics in amateur players. *Scand J Med Sci Sports* 2006; 16: 102–110.
 52. Sáez de Villarreal E, Suarez-Arrones L, Requena B, et al. Effects of plyometric and sprint training on physical and technical skill performance in adolescent soccer players. *J Strength Cond Res* 2015; 29: 1894–1903.
 53. Siegler J, Gaskill S and Ruby B. Changes evaluated in soccer-specific power endurance either with or without a 10-week, in-season, intermittent, high-intensity training protocol. *J Strength Cond Res* 2003; 17: 379–387.
 54. Singh A, Kulkarni K, Shenoy S, et al. Effect of 6 weeks of preseason concurrent muscular strength and plyometric training in professional soccer players. *J Postgrad Med Educ Res* 2014; 48: 27–32.
 55. Wong P, Chaouachi A, Chamari K, et al. Effect of pre-season concurrent muscular strength and high-intensity interval training in professional soccer players. *J Strength Cond Res* 2010; 24: 653–660.
 56. Silva J, Nassis G and Rebelo A. Strength training in soccer with a specific focus on highly trained players. *Sport Med Open* 2015; 1: 1.
 57. Moseley AM, Herbert RD, Sherrington C, et al. Evidence for physiotherapy practice: a survey of the Physiotherapy Evidence Database (PEDro). *Aust J Physiother* 2002; 48: 43–49.
 58. Buchheit M, Mendez-Villanueva A, Simpson BM, et al. Repeated-sprint sequences during youth soccer matches. *Int J Sports Med* 2010; 31: 709–716.
 59. Cohen J. *Statistical power analysis for the behavioral sciences*, 2nd ed. Hillsdale: Lawrence Erlbaum, 1988.
 60. Higgins JP, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *Br Med J* 2003; 327: 557–560.
 61. Markovic G. Does plyometric training improve vertical jump height? A meta-analytical review. *Br J Sports Med* 2007; 41: 349–355.
 62. Buchheit M, Mendez-Villanueva A, Delhomel G, et al. Improving repeated sprint ability in young elite soccer players: repeated shuttle sprints vs. explosive strength training. *J Strength Cond Res* 2010; 24: 2715–2722.
 63. González-Badillo JJ, Rodríguez-Rosell D, Sánchez-Medina L, et al. Maximal intended velocity training induces greater gains in bench press performance than deliberately slower half-velocity training. *Eur J Sport Sci* 2014; 18: 772–781.
 64. Seitz LB, Reyes A, Tran TT, et al. Increases in lower-body strength transfer positively to sprint performance: a systematic review with meta-analysis. *Sports Med* 2014; 44: 1693–1702.
 65. Bishop D, Lawrence S and Spencer M. Predictors of repeated-sprint ability in elite female hockey players. *J Sci Med Sport* 2003; 6: 199–209.
 66. Bishop D, Girard O and Mendez-Villanueva A. Repeated-sprint ability – Part II: recommendations for training. *Sports Med* 2011; 41: 741–756.
 67. Buchheit M, Millet GP, Parisy A, et al. Supramaximal training and postexercise parasympathetic reactivation in adolescents. *Med Sci Sports Exerc* 2008; 40: 362–371.
 68. Mujika I, Santisteban J and Castagna C. In-season effect of short-term sprint and power training programs on elite junior soccer players. *J Strength Cond Res* 2009; 23: 2581–2587.
 69. Loturco I, Nakamura FY, Kobal R, et al. Training for power and speed: effects of increasing or decreasing jump-squat velocity in elite young soccer players. *J Strength Cond Res* 2015; 29: 2771–2779.
 70. Kobal R, Loturco I, Barroso R, et al. Effects of different combinations of strength, power, and plyometric training on the physical performance of elite young soccer players. *J Strength Cond Res* 2016. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27548790>.